18.6.2 DNA Replication

DNA replication is one of the basic process that occurs within a cell. DNA molecule produces exact copies of its own structure during replication process. The two strands of a DNA molecule have complementary base pairs, the nucleotides of each strand provide the information needed to produce its new strand. The two resulting daughter cells contain exactly the same genetic information as the parent cell. DNA replication involves the following steps

Origin of replication

The specific points on the DNA, where the replication begins, is the **site of origin** of replication. The two strands open and separate at this point forming the **replication fork**.

Unwinding of DNA molecule

The enzyme called **helicase**, bind to the origin of replication site. Helicase separates the two strands of the DNA. The enzyme called **topoisomerase** separates the double helix above the replication fork and removes the twists formed during the unwinding process. Each of the separated DNA strands function as a template.

Formation of RNA primer

An RNA primer is a short segment of RNA nucleotides. The primer is synthesized by the DNA template close to the origin of replication site.

Synthesis of new complementary strand from the parent strand

After the formation of RNA primer, nucleotides are added with the help of an enzyme **DNA polymerase** and a new complementary strand of DNA is formed from each of the parent strand. The synthesis is unidirectional.

In one strand, the daughter strand is synthesized as a continuous strand which is called **leading strand.** In the other strand, short segments of DNA are synthesized. This strand is called **lagging strand.** The short segments of DNA are called **Okazaki fragments**. The fragments are joined together by the enzyme, **DNA ligase**.

The replication stops when the replication fork of the two sides meet at a site called **terminus**, which is situated opposite to origin of replication site

18.6.3 Significance of DNA

- It is responsible for the transmission of hereditary information from one generation to next generation.

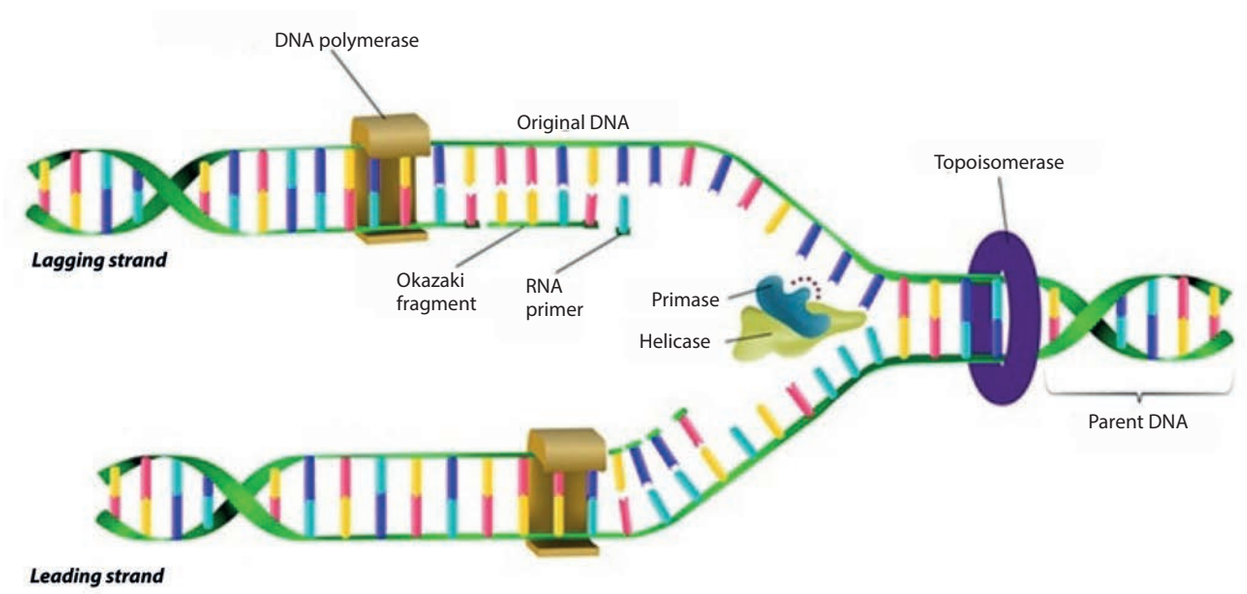


Figure 18.8 Replication of DNA

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